

Problem-Solution Fit

Date	05-07-2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] 1. Farmers 2. Agricultural Officers 3. Agricultural Students 4. Farm Consultants	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] • Limited technical knowledge • Budget constraints • Limited internet access in rural areas • Limited access to agricultural experts	5. AVAILABLE SOLUTIONS PROS & CONS [AS] raditional Farming: Easy to follow but less accurate. Expert Advice: Helpful but not always available. Online Information: Easily accessible but may not be location-specific.
2. PROBLEMS / PAINS + ITS FREQUENCY [PR] • Wrong crop selection (High)	9. PROBLEM ROOT / CAUSE [RC] • Lack of scientific crop selection	7. BEHAVIOR + ITS INTENSITY [BE] • Frequently checks weather forecasts

- Incorrect fertilizer usage (Medium)
- Unpredictable weather conditions (High)

- Unpredictable climate
- Improper soil nutrient management
- Limited access to expert guidance

- Consults local farmers
- Uses fertilizers based on experience
- Intensity: High during crop season

3. TRIGGERS TO ACT [TR]

- Crop failure
- Poor harvest
- Increasing farming costs
- Need for higher productivity

4. EMOTIONS

BEFORE / AFTER [EM]

Before: Worried, Confused, Uncertain

After: Confident, Satisfied, Hopeful

10. YOUR SOLUTION [SL]

OptiCrop uses Machine Learning to analyze soil nutrients (N, P, K), temperature, humidity, pH, and rainfall to recommend the most suitable crop. The system improves farming decisions, increases crop yield, reduces resource wastage, and supports sustainable agriculture.

8. CHANNELS OF BEHAVIOR [CH]

Online: Mobile App, Website, Agricultural Portals, Weather Apps

OFFLINE:- Local Agriculture Office, Farmer Meetings, Agricultural Experts